

Foundation Models and Large Language Models

Generative AI

Module 1 – Lesson 2

Today's activities

- Traditional ML
- Foundation models and LLMs
- Transformers Architecture
- Challenges and Limitations of LLMs



Review: Traditional ML (1/2)

- Trained on task-specific data
 - Models specialized for the task
- Training typically starts from scratch
- Several model choices
 - Tree-based models, linear models, neural networks, etc.

Review: Traditional ML (2/2)

- Optimized for one task
 - Classification, regression, clustering
- Challenging to adapt to another similar task

Review: Machine learning terminology

ML: Train a computer to recognize **patterns** in historical data to make **predictions** on new data

The ML algorithm takes a random **function** and refines it until the **features** predict the correct **label**.

f

model

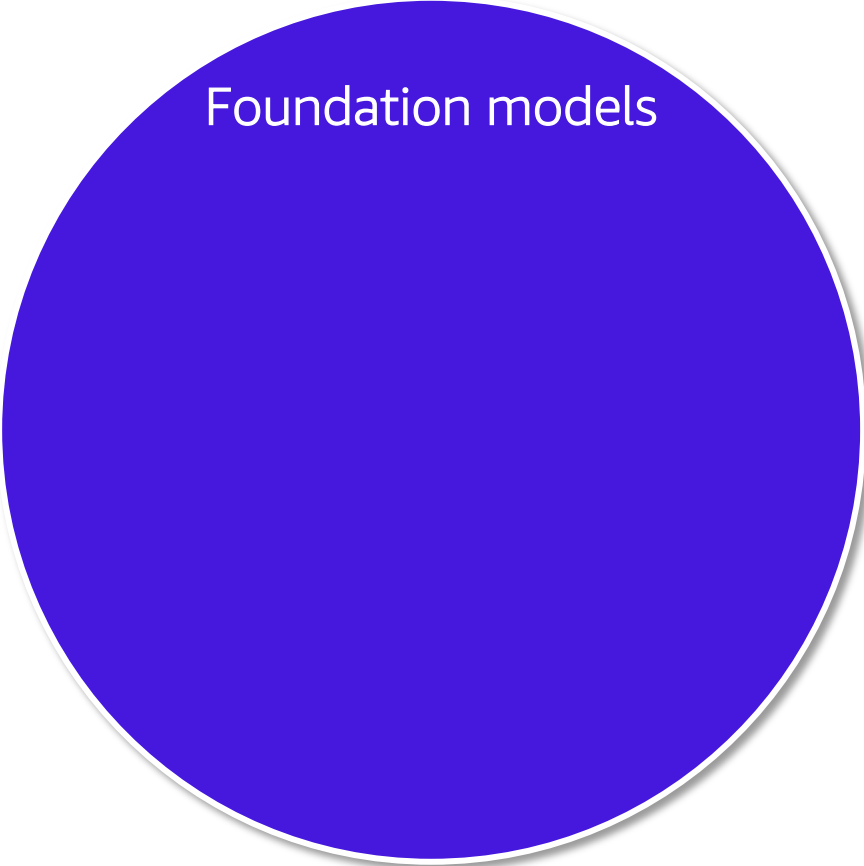
features

Number of logins	Watched Prime Video	...	Number of purchases
120	Yes	...	4
1	No	...	0
219	No	...	12
57	Yes	...	2
0	No	...	0
23	Yes	...	1

label

Prime signup after free trial
Yes
No
Yes
Yes
No
No

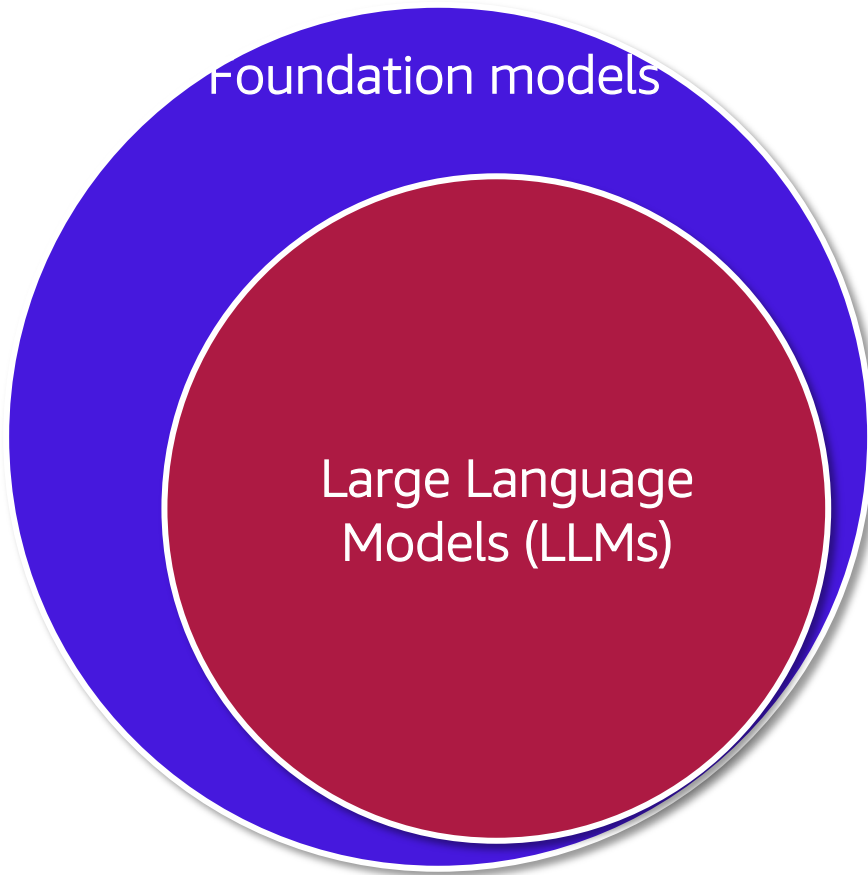
Review: Foundation models



Foundation models

- Large ML models that are **pre-trained** with **vast amounts of data**. These can be **adapted** to more specialized tasks
- Can be trained on any kind of data
 - Text
 - Images
 - Video
 - Audio
 - Etc...

Review: Large Language Models (LLMs)

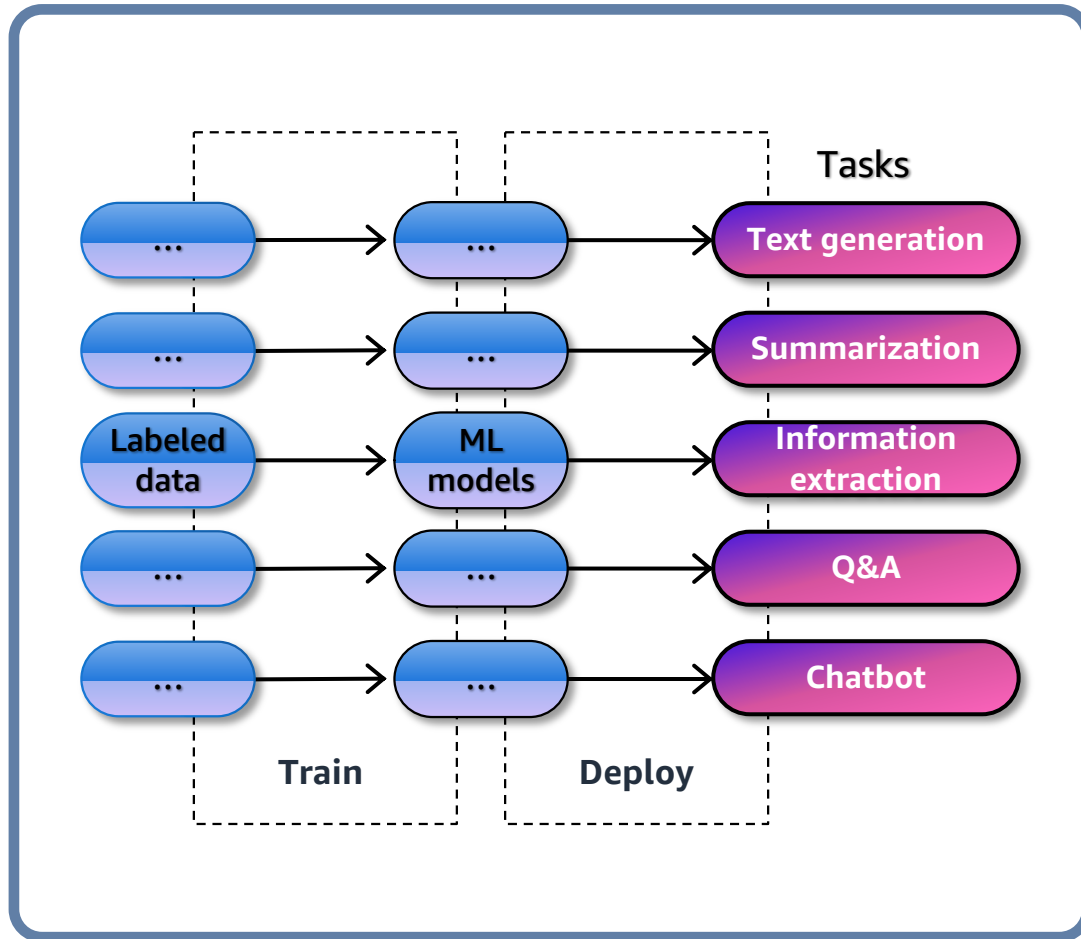


- Foundation models trained on **text**
- Large ML models that learn the **probabilities of words** being used in certain contexts
- **Training task:** Learn to predict the missing word in a text sequence

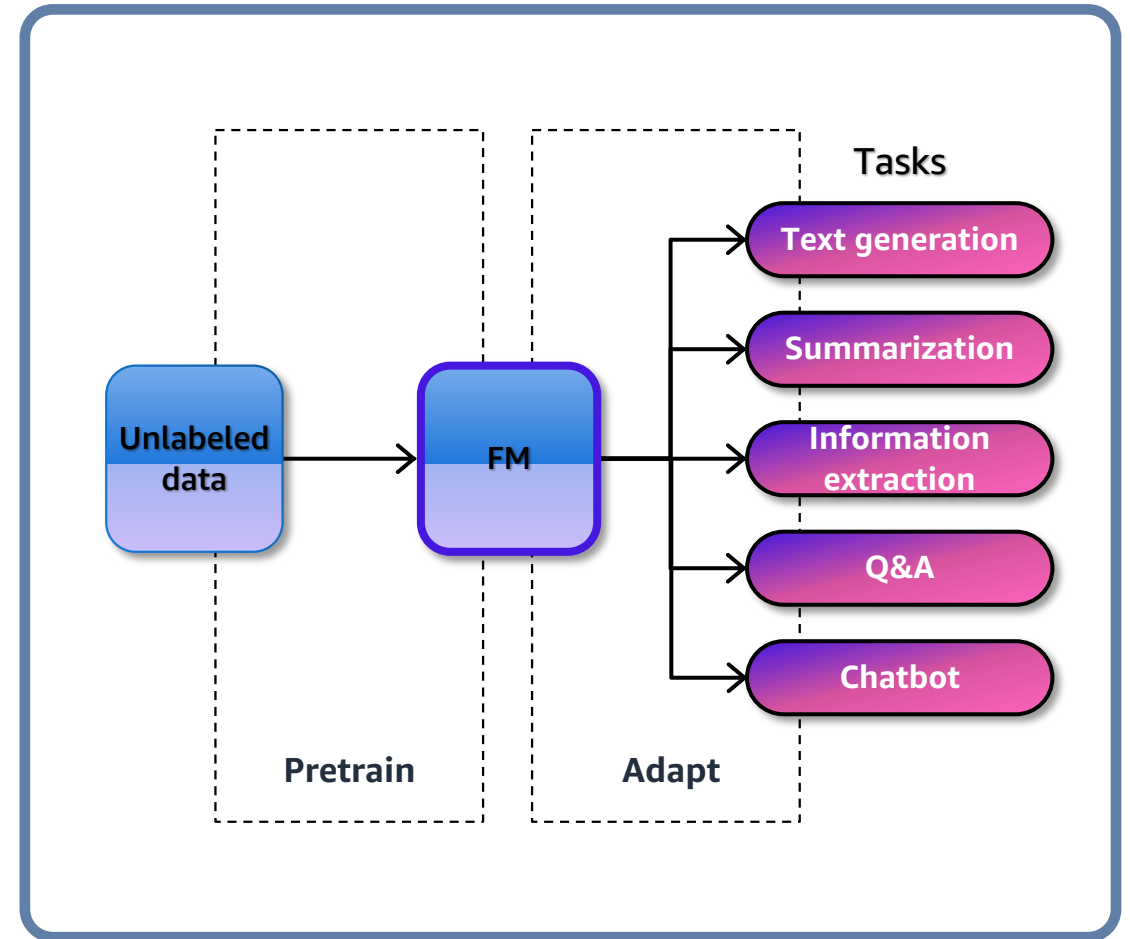
"The weather has been cloudy for the last two days. Most likely it will be ____ tomorrow."

cloudy? sunny? foggy?

Foundation models (FMs)



Traditional ML models



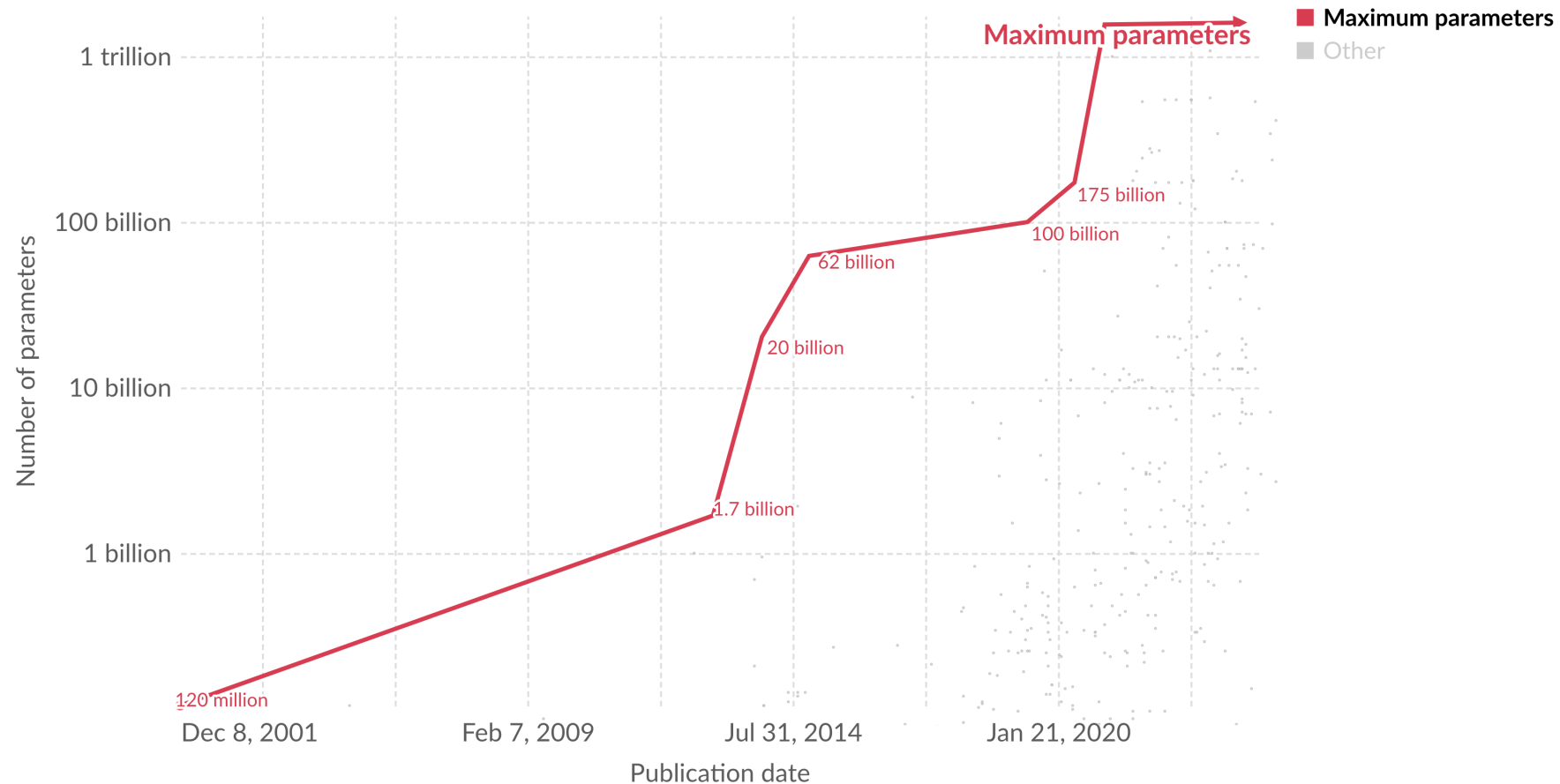
Foundation models

The growth of LLMs

Parameters in notable artificial intelligence systems

Our World
in Data

Parameters are variables in an AI system whose values are adjusted during training to establish how input data gets transformed into the desired output; for example, the connection weights in an artificial neural network.



Source: [ourworldindata](https://ourworldindata.org)

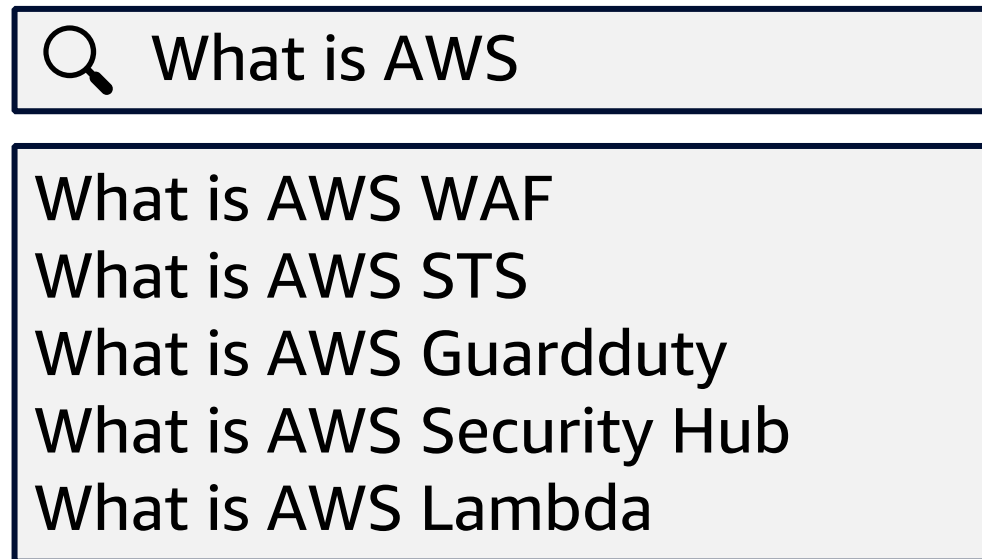
Transformer Networks

Sequence-to-sequence models

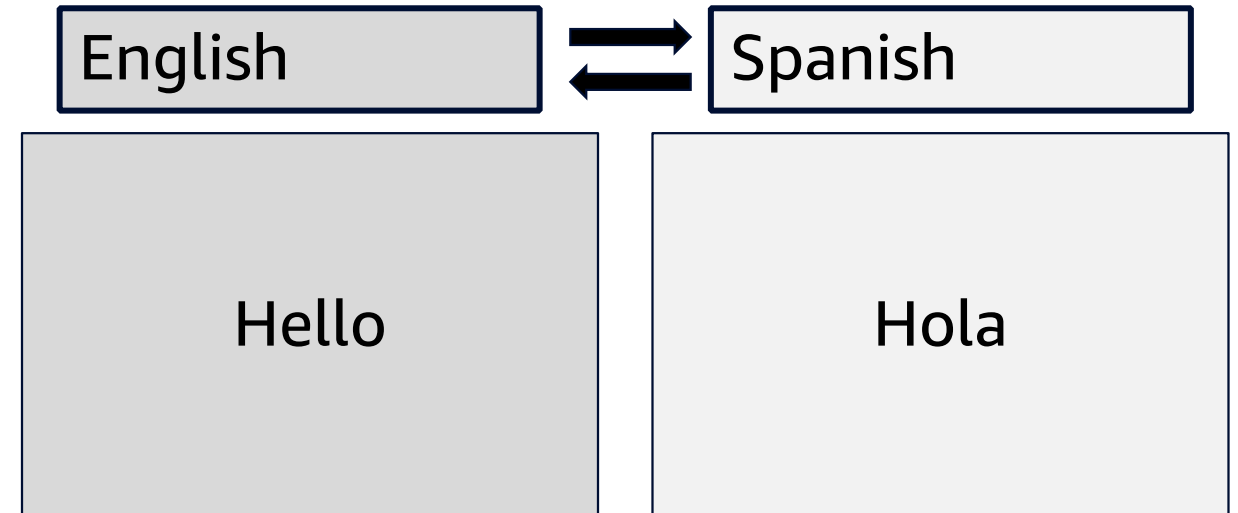
Use input sequences to produce output sequences.

- Input and output are **both variable-length** sequences
- Examples:

Auto-completion

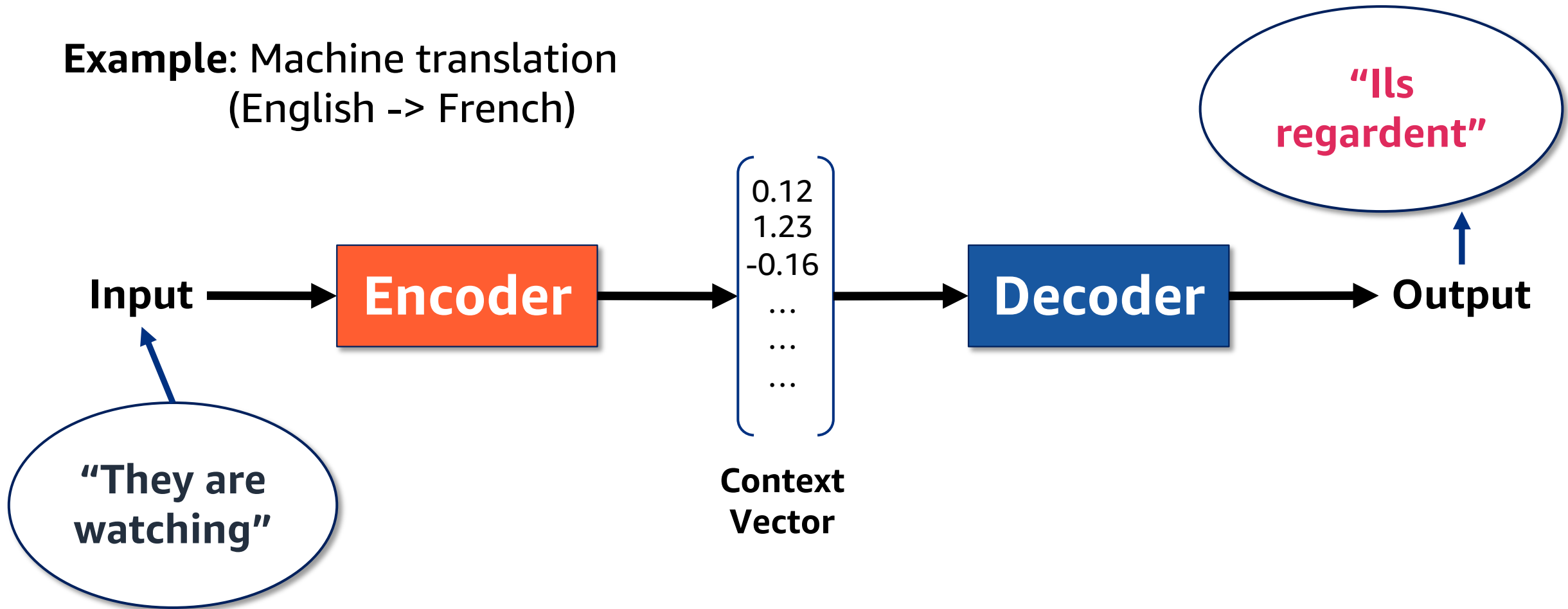


Machine translation



Encoder-Decoder Architecture

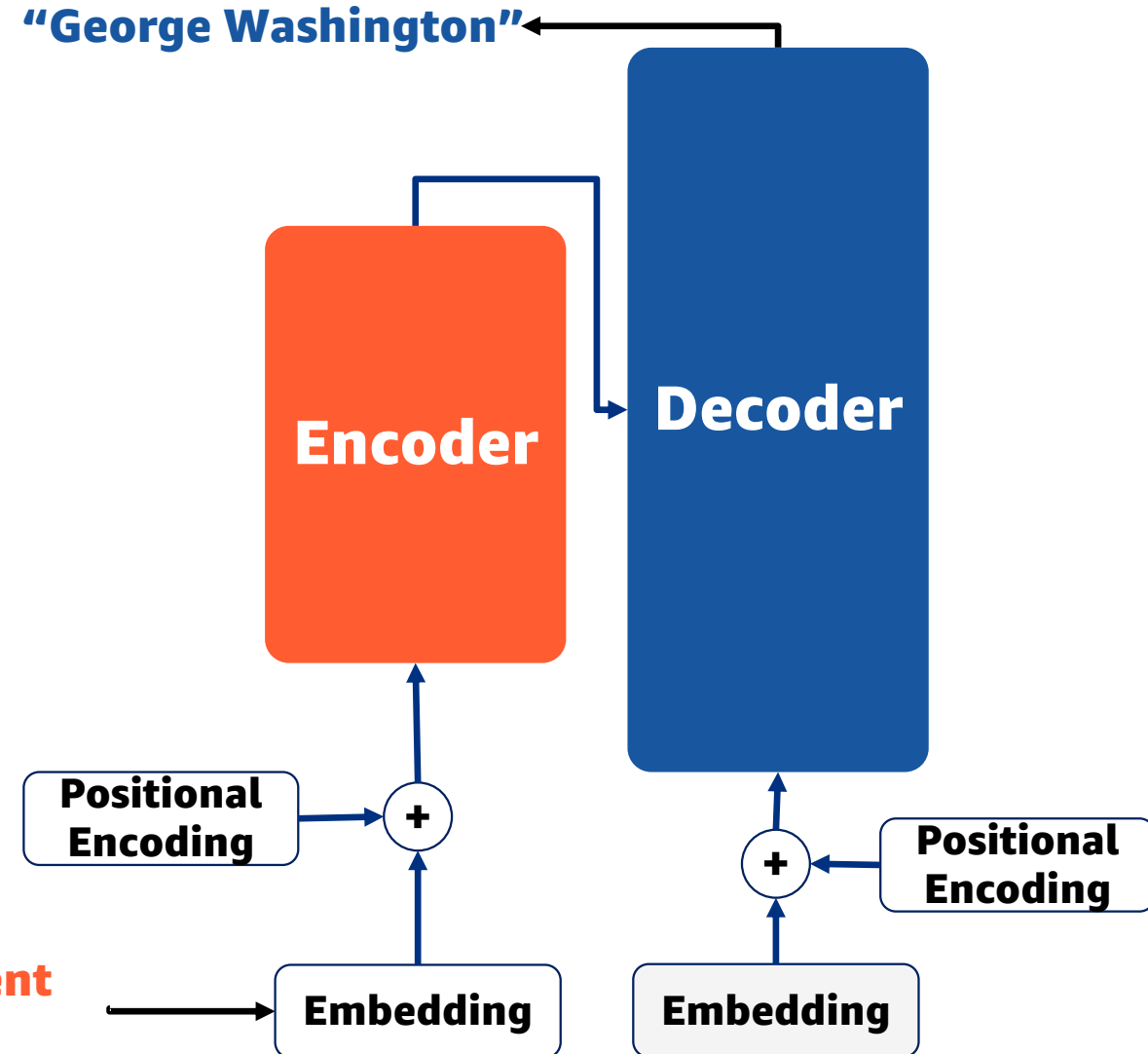
Example: Machine translation
(English -> French)



Transformers

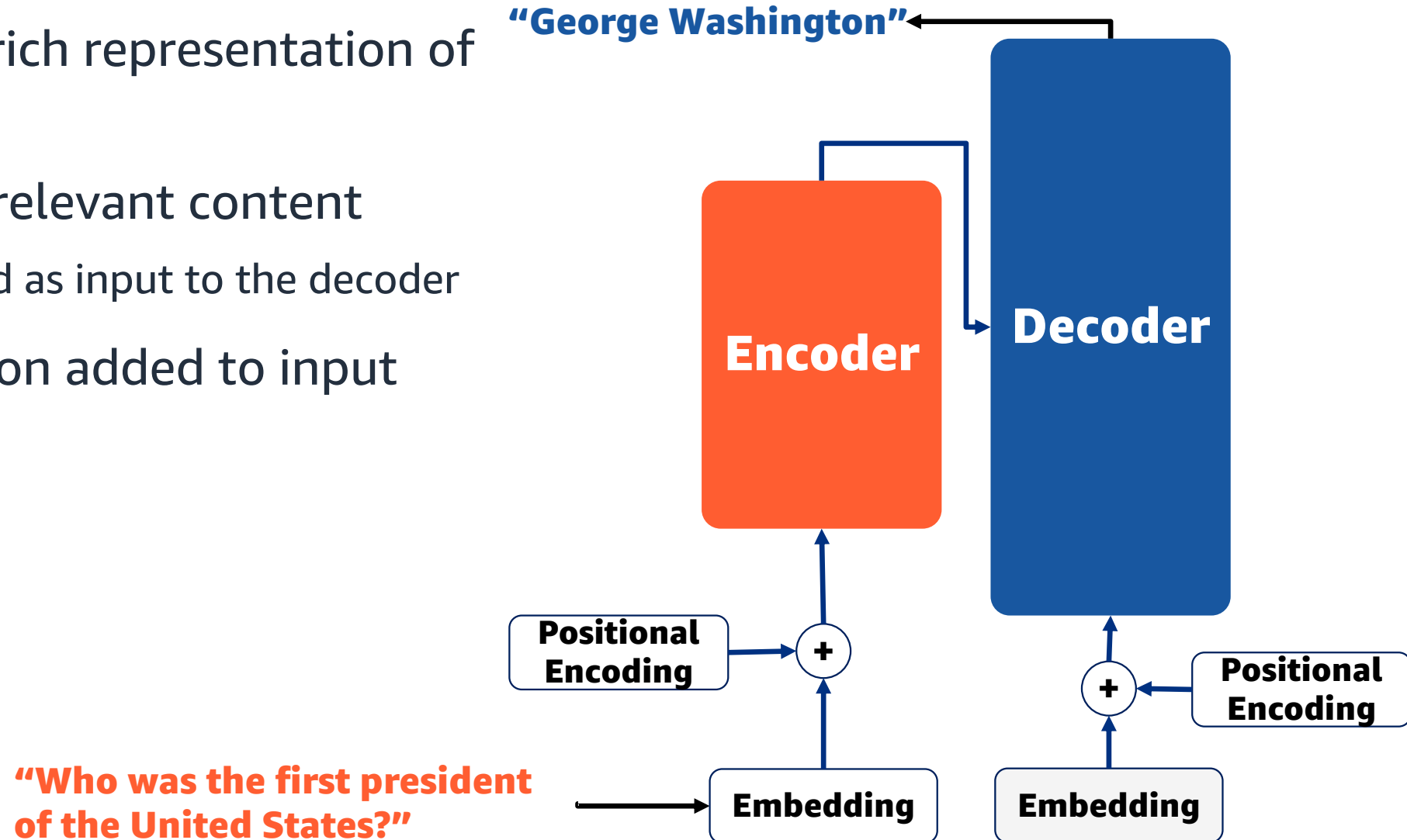
- Developed as an encoder-decoder model
- Uses the **attention mechanism**
 - Rich semantic and syntactic representation
 - Long context memory
- Parallel processing of inputs
- Distributed inference

“Who was the first president of the United States?”



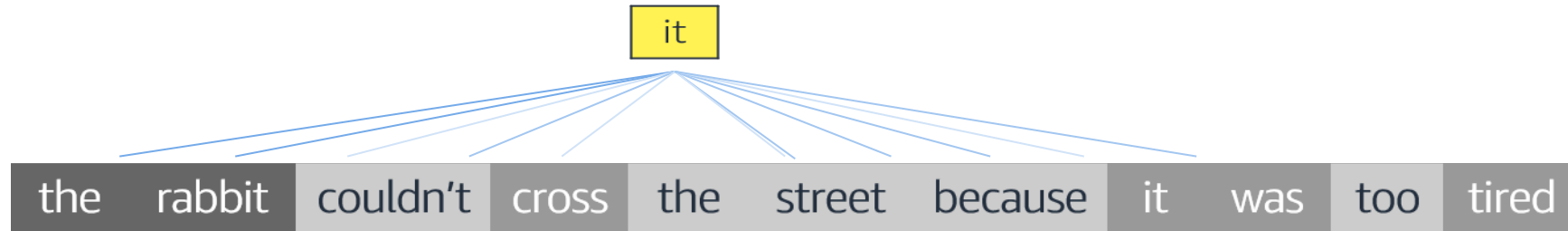
Transformers

- **Encoder** generates rich representation of input
- **Decoder** generates relevant content
 - Output is again passed as input to the decoder
- Positional information added to input



Attention Mechanism

- Attention helps models **associate each word** with **other words** in the sentence



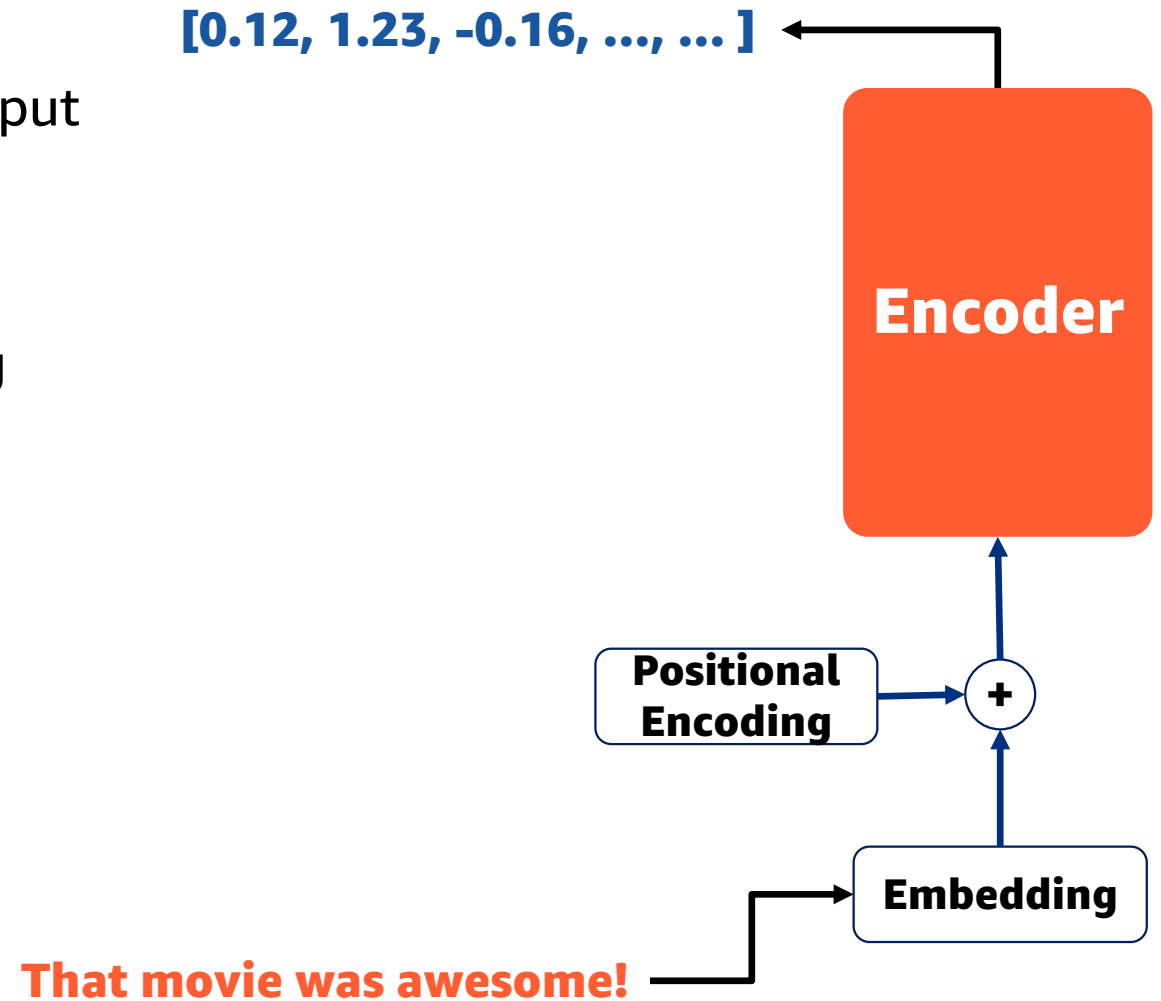
■ High attention
■ Mid attention
■ Low attention

Attention Mechanism

- Allows the model to access all the words/tokens in the observable input
- Indicates which tokens are most relevant for the next prediction
- Both the encoder and decoder use the attention mechanism
 - **Multi-headed attention module**

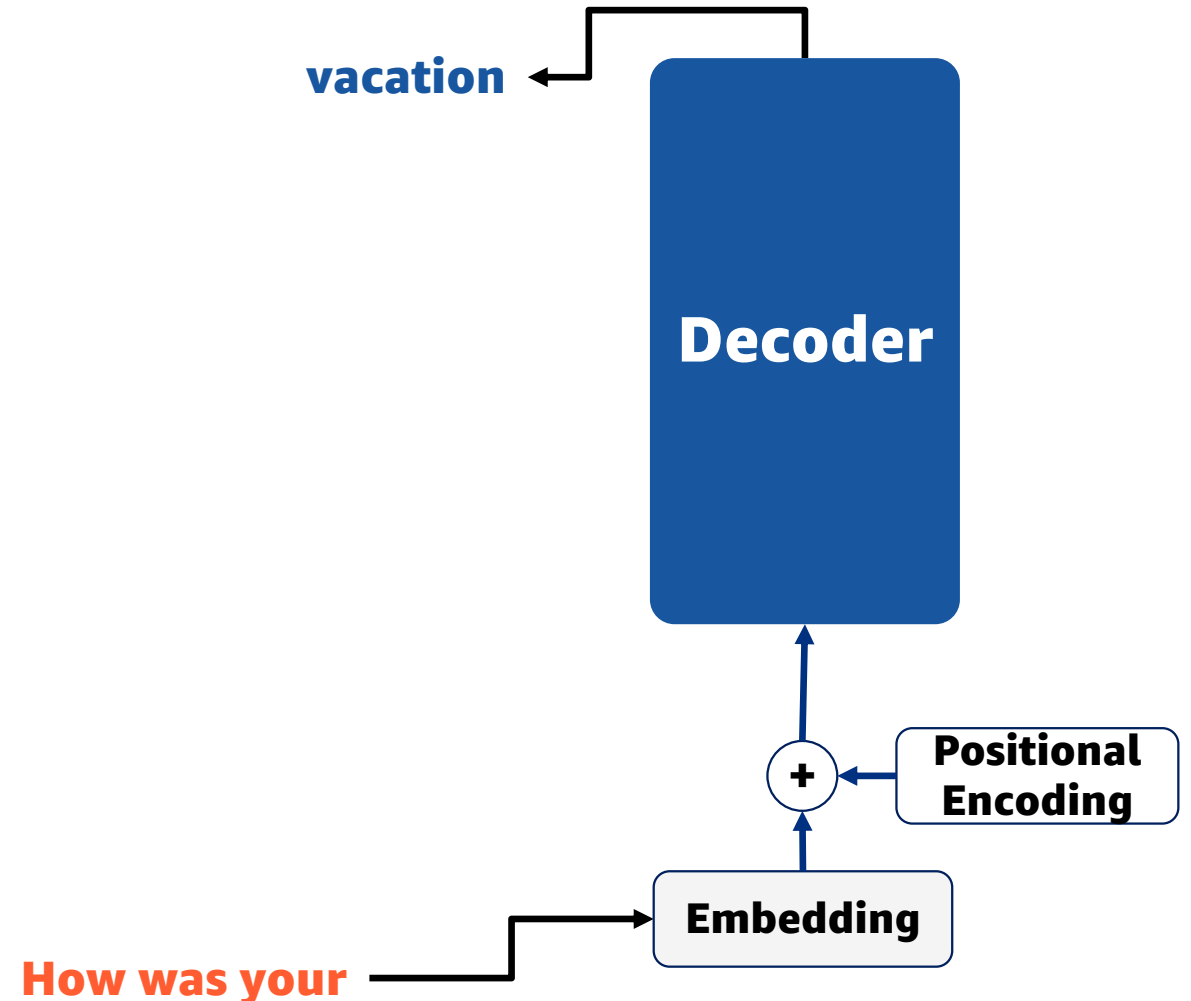
Encoder-only Architecture

- Only uses the encoder block
- Attention layers access all the words in the input sentence
 - **Bi-directional attention**
- Suitable for Natural Language Understanding (NLU) tasks
 - Sentence classification
 - Named entity recognition (NER)
 - Text extraction
- Examples:
 - BERT Models
 - ELECTRA



Decoder-only Architecture

- Only uses the decoder block
- Only access words that came before
 - **Auto-regressive models**
- Generate next token based on input
- Suitable for text generation tasks
- Examples:
 - GPT Models by OpenAI
 - Llama models by Meta
 - Claude models by Anthropic



Transformers: Summary

- **State-of-the art** deep learning architecture
- Propelled the growth and adoption of **Generative AI**
- Transformers can process the input data in **parallel**
- Self-attention mechanism capture interdependencies between all words, regardless of position
- Typically undergo **self-supervised learning**
 - Labels generated automatically from unlabeled data
 - No need to curate labelled data
- Increasingly popular with Computer Vision (CV), Audio, Reinforcement Learning (RL) tasks, and multi-modal applications

Transformers for all!

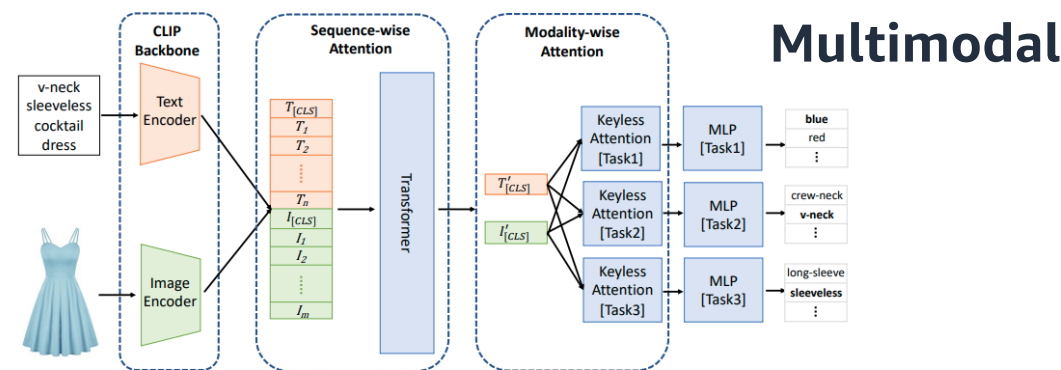
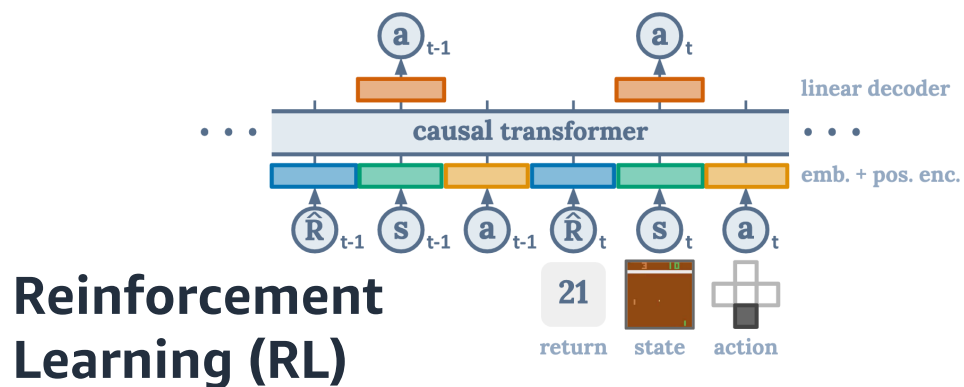
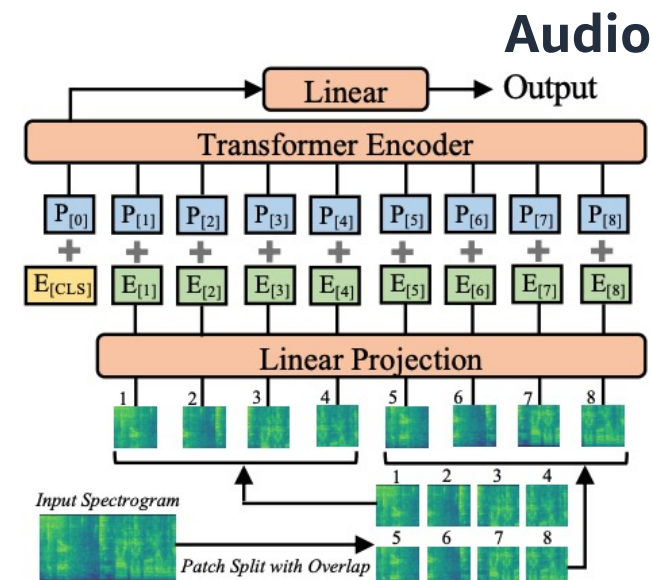
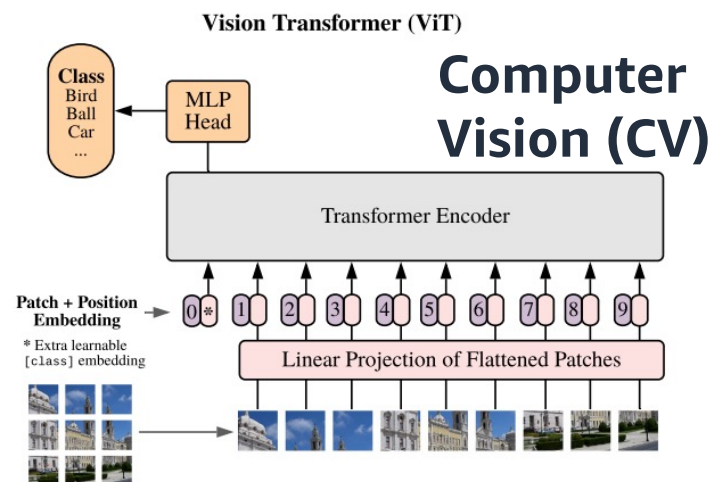
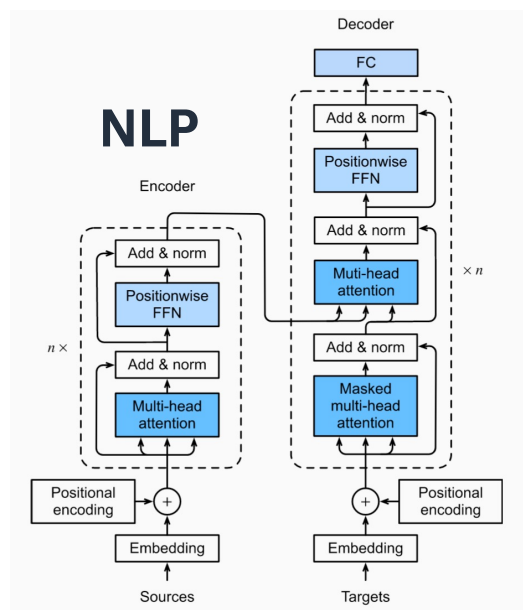


Figure 1: The pipeline of our proposed CMA-CLIP.

Challenges and Limitations of LLMs

Challenges and limitations of LLMs (1/3)

- **Reliability and bias**
 - Knowledge limited to the training data
 - Inability to discern false information or bias
- **Context window**
 - Model's attention limited to the context window
 - Inputs exceeding the context window length are invisible to the model
 - For instance, Titan Premier had a limit of 30,000 tokens at the time of release

Challenges and limitations of LLMs (2/3)

- **Potential copyright infringement issue**
 - Training data might contain sensitive or copywrite data
 - May generate content similar to someone's intellectual or creative property
 - Have ethical and legal implications
- **Create and propagate misinformation**
 - May generate personal or sensitive data that could be used to identify and harm others
 - Create and propagate misinformation about individuals, groups, organizations, etc.

Challenges and limitations of LLMs (3/3)

- **System costs**

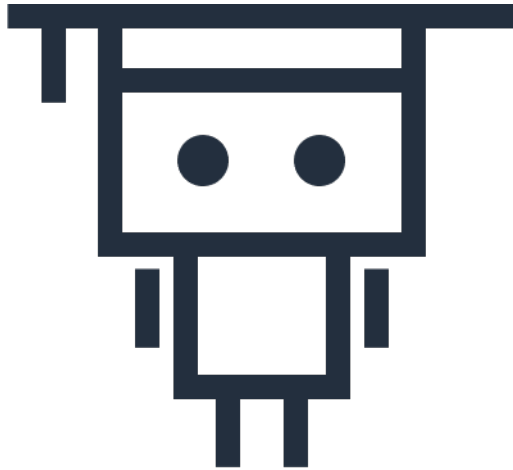
- Requires significant investment in the form of computer systems, human capital (engineers, researchers, scientists, etc.), and power
- Models with >100 billion parameters can have a total project cost of over \$100 million

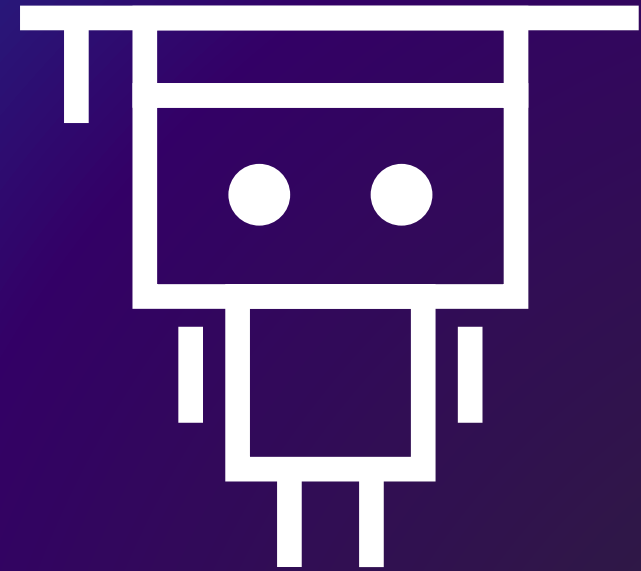
- **Environmental Impact**

- LLMs need a lot of power and leave behind large carbon footprints
- According to a study, CO₂ emissions from training 5B models on GPU is roughly equivalent to a trans-American flight

Next lesson

- This lesson covered foundation modes and LLMs
- In the next lesson, you will learn about prompt engineering and a few techniques to improve the quality of the model's response through prompting





Thank you!